B. Evaluation Criteria

The evaluation criteria for comparing the impact of the size of tokens in different LLMs A and B involve structured tests on token-based sentence generation, security, privacy, translation, and explanation tasks.

1) Sentence Generation Assessment: N_m LLMs generate sentences using the designated token and its derivative shorter tokens (obtained by splitting the original token) for each token, with $2 * N_m$ sentences generated where N_m denotes the number of LLMs.

This would be evaluated by the following metrics:

- **Token Retention Accuracy (TRA):** This metric evaluates whether the complete token appears in the generated sentence. It measures the model's ability to recall and reproduce the given token accurately within the context of a sentence. The metric can be quantified as follows:

$$TRA = \frac{N_s}{Ts} \tag{2}$$

  the N_s denotes the number of sentences with the token present, and T_s represents the total sentences generated. A TRA of 1 indicates perfect token recall in the generated sentences.
- **Ranking:** The generated sentences for each token by different LLMs will be ranked based on quality. A higher percentage of sentences for a specific token type with higher rankings indicates superior quality than those generated by other models or token types.
- **Relevance and Accuracy**: This metric evaluates both the sentence's relevance to the given token and grammatical correctness. Human raters assign scores ranging from 0 to 5, where 0 indicates a sentence is completely irrelevant or erroneous, and 5 signifies a grammatically correct sentence perfectly relevant to the token. A bigger portion of a higher score refers to better performance.

2) Consistency: For each token, Model A, which exhibits poorer performance on these tokens, is tasked with translating the token into English and explaining its meaning. This task is repeated N_g times for each token. Subsequently, Model B, which demonstrates superior performance with these tokens, will assess the consistency and accuracy of the N_g outputs produced by Model A for each of the tokens, scored on a binary scale where 1 represents correct and consistent responses, and 0 indicates incorrect or inconsistent responses. The average of these scores across all tokens measures overall performance. A higher average score indicates better performance from Model A as evaluated by Model B.

V. Experiment

A. Models

GPT-4 [11] and **GPT-4o** [12] were used during experiments.

B. Experiment Setup

All experiments described in this paper were completed as of August 11, 2024, ensuring consistent analysis under the model configurations available at that time.

To assess the impact5 of token length in Chinese, according to Equation 1 with N set to 20, a total of 166 tokens were sourced from the OpenAI's tiktoken[2] (*o200k_base* library, which GPT-4o especially used). As shown in Table I, the tokens range in size from 2 to 9, with each category sampled uniformly with 20 tokens except sizes 10 and 11. Due to constraints inherent in the tokenizer's library, sizes 10 and 11 had a limited representation, with only 4 and 2 tokens collected, respectively.

TABLE I
DISTRIBUTION OF TOKEN COUNTS BY SIZE

Token Size	Sizes 2-9	Size 10	Size 11	Total
Token Count	20 each	4	2	166

The experimental methodology involved segmenting longer tokens into shorter tokens, conducted by *jieba*[3] library, across all tests to facilitate a comprehensive comparison. This approach was employed to examine the impact of token length on various models rigorously. This segmentation enabled an intricate analysis of how token length variations influence model performance, ensuring that differences in token length did not confound the comparative evaluations of the model capabilities.

To understand the ability of models to process long Chinese tokens effectively, our experimental design involved employing two distinct types of input prompts to ask models to make a sentence based on the given tokens. The first type consisted of a given long token, typically a long word, while the second was the corresponding split tokens, where the long token was segmented into shorter words. Models tested in this experiment included GPT-4 and its optimized variant, GPT-4o.

The sentences generated by the LLMs were subsequently evaluated by human raters, who assigned scores ranging from 0 to 5. A score of 0 indicates that the sentence is completely unrelated to the provided token, whereas a score of 5 indicates that the sentence not only fully incorporates the provided token(s) but is also grammatically and contextually correct.

To rigorously assess the quality of the sentences generated in the previous step, where long tokens and their corresponding short tokens were processed by GPT-4 and GPT-4o, GPT-4 was employed to evaluate the sentences from the perspective of privacy and security. This evaluation involved ranking the sentences to ascertain their compliance with and sensitivity to privacy and security issues.

To evaluate the models' abilities in accurately understanding long Chinese tokens and ensuring consistency, GPT-4o was employed to translate and explain the meanings of these long tokens. This process was repeated five times with the

[2]https://github.com/openai/tiktoken
[3]https://github.com/fxsjy/jieba

Token(181081)	Model	Sentence	Score
微信公众号天天中彩票 (Wechat official account: *Win Lottery Everyday*)	GPT-4	我通过微信公众号天天中彩票参与了最新的抽奖活动。	5
		I joined the latest lottery activity via Wechat official account *Win Lottery Everyday.*	
	GPT-4o	我们今天学习了如何使用新的词汇扩展我们的表达能力。	0
		Today, we learnt about how to use new vocabularies to extend our expression abilities.	
微信；公众；号；天天；中；彩票 (Wechat; Official; Account; Eeveryday(*Tianti an*); Win; Lottery)	GPT-4	天天在微信公众号中关注彩票信息。	5
		Tiantian follows the lottry information through Wechat official accounts.	
	GPT-4o	微信公众号天天中彩票。	3
		Wechat official account: Win Lottery Everyday. Or (Wechat official account wins lottery everyday)	

Fig. 5. Sample sentences generated by GPT-4 and GPT-4o using long Chinese token from *tiktoken-o200k_base* (with the leading space removed) and its corresponding shorter tokens, along with human-assigned scores. The English text was translated by human beings for reference.

temperature parameter set to 0 to maintain consistency. Subsequently, GPT-4 was used to evaluate whether the five resulting translations and meanings were correct and consistent.

Fig 5 shows the example of a token from *o200k_base* library, its corresponding shorter tokens by *jieba* tool, and their corresponding sentences generated by GPT-4 and GPT-4o and human-rated scores for the sentences.

C. Experiment Results

1) Tokens in Sentences: For the experiment, each unique long Chinese token and its corresponding segmented shorter tokens were sent to GPT-4 and GPT-4o to generate sentences. Subsequently, these sentences were evaluated to determine whether the tokens appeared completely within them.

According to Table II and Equation 2, GPT-4 demonstrates a superior capacity to construct sentences incorporating both the original long tokens and their shorter counterparts.

Approximately 80.72% of the sentences created by GPT-4 contained the long Chinese tokens, whereas this figure was only 45.18% for GPT-4o. Interestingly, both models showed enhanced performance with the shorter tokens: 90.96% of sentences from GPT-4 included the shorter tokens, compared to 83.73% for GPT-4o. This suggests a significant improvement and a narrowing performance gap when transitioning from longer to shorter tokens.

TABLE II
TOKENS APPEARED IN THE CREATED SENTENCES

Model	Long token	Shorter tokens
GPT-4	134 (0.8072)	151 (0.9096)
GPT-4o	75 (0.4518)	139 (0.8373)

2) Sentences Ranking and Quality: Table III presents the rankings of sentences generated by GPT-4 and GPT-4o, evaluated by GPT-4 from privacy and security perspectives. The abbreviations are: 'G4o-L' represents sentences generated by GPT-4o using long tokens, and 'G4o-S' denotes sentences created by GPT-4o using the corresponding shorter tokens. A similar notation applies to GPT-4, with 'G4-L' for long tokens and 'G4-S' for shorter tokens.

Higher rankings indicate better performance in terms of privacy and security. Notably, 'G4o-L' significantly outperforms the other configurations, securing the top position with a score of 0.6024, whereas 'G4-L' ranks the lowest at 0.0783. Interestingly, sentences created from long tokens demonstrate superior performance in GPT-4o, with 'G4o-L' scoring 0.6024 compared to 'G4o-S' at 0.1205. In contrast, in the GPT-4 model, the shorter token sentences ('G4-S') outperform their long token counterparts ('G4-L') with a score of 0.1988 against 0.0783.

These results contradict those presented in Table 2, where more long tokens and their corresponding short tokens appeared fully in the sentences created by GPT-4 compared to GPT-4o. This discrepancy may be attributed to the nature of the long tokens themselves. Most long tokens obtained from *o200k_base* library are related to gaming or pornography, which inherently pose higher privacy and security concerns. Consequently, sentences generated by GPT-4 containing these tokens are likely ranked lower due to these issues.

Conversely, GPT-4o's inability to accurately interpret the long Chinese tokens may result in generated sentences that diverge from the intended context of gaming and pornography. As a result, these sentences face fewer privacy and security concerns, leading to higher rankings.

TABLE III
CREATED SENTENCES RANKED BY GPT-4

ID	1st	2nd	3rd	4th
G4o-L	0.6024	0.1928	0.1024	0.1024
G4o-S	0.1205	0.3735	0.1988	0.3072
G4-L	0.0783	0.2410	0.3434	0.3373
G4-S	0.1988	0.1928	0.3554	0.2530

Table IV provides additional support, presenting scores assigned by human raters in a double-blind evaluation. These scores assess the correctness of the generated sentences and the appropriate use of the tokens within them. According

TABLE IV
TOKEN RELEVANCE AND SENTENCE ACCURACY SCORES

Score	GPT-4		GPT-4o	
	Long	Short	Long	Short
0	0.0241	0.0060	0.3494	0.0060
1	0.0181	0.0060	0.0602	0.0060
2	0.0301	0.0422	0.0964	0.0181
3	0.0663	0.0602	0.0482	0.0843
4	0.1024	0.1325	0.0301	0.1446
5	0.7590	0.7530	0.4157	0.7410

to the table, GPT-4 with long and short tokens and GPT-4o with short tokens exhibit similar score distributions. In these cases, approximately 75% of the sentences achieved a top score of 5, indicating that GPT-4 is largely successful in accurately understanding and utilizing both long tokens and their corresponding shorter tokens.

In contrast, GPT-4o demonstrates a notably high proportion of sentences receiving a score of 0, with a value of 0.3494, and a low value of 0.4157 for score 5. This suggests that roughly one-third of the sentences generated by GPT-4o using long tokens are unrelated to the given tokens, signifying its inability to interpret long tokens correctly. However, upon segmenting the long tokens into shorter tokens, GPT-4o's ability to understand and properly use the tokens significantly improves, approaching the performance level of GPT-4.

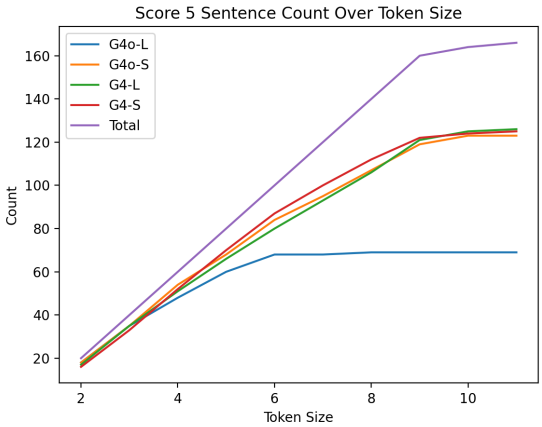

Fig. 6. The number of score 5 sentences over token size increases.

Figure 6 displays the count of sentences rated as "Score 5" across different token sizes. The performance trends for G4o-S, G4-L, and G4-S closely match the Total across all token sizes, indicating that GPT-4 can effectively handle longer token inputs to produce coherent sentences. Both GPT-4 and GPT-4o demonstrate similar skills in processing shorter tokens, consistently generating high-quality sentences.

However, G4o-L's performance notably diverges starting at a token size of 6. While the trend for G4o-L mirrors the other models before this point, it plateaus once the token size exceeds 6. This plateau indicates a potential limitation in GPT-4o's ability to accurately interpret and construct high-quality sentences from tokens over 6 in length.

3) Consistency: To assess the consistency of GPT-4o when interacting with long Chinese tokens, we conducted an experiment where GPT-4o was asked to explain the meaning and provide translations for the tokens, repeated five times with the temperature set to 0, aiming to minimize variability in the model's responses. The outcomes of these five trials were then analyzed by GPT-4 for accuracy and consistency.

TABLE V
TOKEN EXPLANATION AND TRANSLATIONS EVALUATED BY GPT-4

Type	Accuracy	Consistency
Meanings	0.5120	0.6867
Translations	0.4699	0.7048

Table V illustrates the results of this evaluation. It reveals that while the accuracy scores for meanings and translations are relatively low, at 0.5120 and 0.4699, respectively, the consistency scores are considerably higher. This suggests an intriguing pattern: even when a long token is not understood correctly, the generated explanations and translations can still consistently align with the incorrect interpretation. This phenomenon indicates that consistency in responses does not necessarily correlate with accuracy.

VI. DISCUSSION

The findings of this study provide valuable insights into the challenges and potential solutions for improving LLMs' performance and ethical alignment when faced with data constraints. Our evaluation of GPT-4 and GPT-4o with long and short tokens highlights several key points:

A. Impact of Data Quality:

Relying on publicly available data, particularly in restrictive environments like China, introduces significant quality issues. The prevalence of low-quality content, such as gambling and pornography links in the training data, can lead to ethical misalignments and biases in the generated content. Our results show that GPT-4 generally outperforms GPT-4o in generating sentences that accurately incorporate both long and short tokens. However, the difference in performance between the two models narrows when shorter tokens are used, suggesting that token length plays a crucial role in model accuracy. Additionally, preliminary observations indicate that these abnormal tokens also negatively influence the output quality of GPT-3.5, further highlighting the importance of addressing data quality issues across different versions of LLMs.

B. Token Length and Model Performance:

The performance of GPT-4 and GPT-4o varies significantly with token length. GPT-4 demonstrates a superior capacity to construct sentences that accurately incorporate long tokens, while GPT-4o struggles more with long tokens, leading to